

Estimations of Quantitative Features

15/11/2023

Description

This file contains the code for estimating the feature values (TTR, unigram entropy, entropy rate, repetition rate) per sign string in the files with randomizations.

Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`. For the Hrate package this is different, since it comes from github. The devtools library needs to be installed, and then the `install_github()` function is used.

```
library(stringr)
library(entropy)
library(gsubfn)
```

```
## Loading required package: proto
```

```
#library(devtools)
#install_github("dimalik/Hrate")
library(Hrate)
```

Load files

Load the files with the randomized strings and put them into one data frame.

```
aur.sample <- read.csv("~/Github/UpperPalCombinatorics/data/aurignacian/swabianJura_randomized.csv")
# add information about period
aur.sample$period <- rep("aurignacian", nrow(aur.sample))
mag.sample <- read.csv("~/Github/UpperPalCombinatorics/data/magdalenian/StGermain_randomized.csv")
# add information about period
mag.sample$period <- rep("magdalenian", nrow(mag.sample))
# merge in one data frame
all.samples <- rbind(aur.sample, mag.sample)
```

Estimations per file

```
# set counter
counter = 0
# initialize dataframe to append results to
estimations.df <- data.frame(id = numeric(0), period = character(0),
                             coding_sample = character(0), num.char = numeric(0),
                             huni.chars = numeric(0), hrate.chars = numeric(0),
```

```

ttr.chars = numeric(0), rm.chars = numeric(0))

# start time
start_time <- Sys.time()
for (i in 1:nrow(all.samples)){
  try({ # if the processing fails for a certain row, there will be no output for this row,
    # but the try() function allows the loop to keep running

    # collect info from row
    id <- all.samples$id[i]
    period <- all.samples$period[i]
    coding_sample <- all.samples$coding_sample[i]
    num.char <- nchar(all.samples$coding_sample[i])

    # split strings into UTF-8 characters
    chars <- unlist(strsplit(coding_sample, ""))
    # preprocess text with bag-of-words normalization
    # note: this is an important step! Without it, entropy rate estimation will fail
    # to yield realistic results due to issues with processing special characters.
    chars <- PreprocessText(chars)

    # unigram entropy estimation
    # calculate unigram entropy for characters
    chars.df <- as.data.frame(table(chars))
    # print(chars.df)
    huni.chars <- entropy(chars.df$Freq, method = "ML", unit = "log2")

    # entropy rate estimation
    # the values chosen for max.length and every.word will crucially
    # impact processing time. In case of "max.length = NULL" the length is n units
    # here use the try() function since entropy rate estimation can fail when particular
    # special characters are in the strings
    hrate.chars <- try(get.estimate(text = chars, every.word = 1, max.length = NULL))
    if (class(hrate.chars) == "numeric") {
      hrate.chars <- hrate.chars
    } else {
      hrate.chars = "NA"
    }

    # calculate type-token ratio (ttr)
    ttr.chars <- nrow(chars.df)/sum(chars.df$Freq)

    # calculate repetition measure according to Sproat (2014)
    # the maximum number of adjacent repetitions "R" possible is the sum of frequency counts minus 1.
    R <- sum(chars.df$Freq)-1
    # calculate the number of adjacent repetitions "r"
    r = 0
    if (length(chars) > 1){
      for (i in 1:(length(chars)-1)){
        if (chars[i] == chars[i+1]){
          r = r + 1
        } else {
          r = r + 0
        }
      }
    }
  })
}

```

```

    }
  }
  # calculate the repetition measure
  rm.chars <- r/R
} else {
  rm.chars <- "NA"
}

# append results to dataframe
local.df <- data.frame(id, period, coding_sample, num.char,
                      huni.chars, hrate.chars, ttr.chars, rm.chars)
estimations.df <- rbind(estimations.df, local.df)
}) # end of try()
# counter
counter <- counter + 1
print(counter)
}

```

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end_time <- Sys.time()
end_time - start_time

## Time difference of 3.254007 secs
#estimations.df
```

Safe outputs to file

```
write.csv(estimations.df, "~/Github/UpperPalCombinatorics/results/features.csv", row.names = F)
```